

Overview

Olive Turtle is our next prototype deployment — code named Olive Turtle — in which we will develop and deploy our navigation stack for the first time. It is an autonomous sailing craft targeting a Mediterranean open-sea crossing via a sequence of pre-loaded GPS waypoints. The autonomy stack is governed by three principles: **determinism** (predictable, testable behaviour on low-power hardware), **simplicity** (minimal moving parts in both hardware and software), and **fault-tolerance** (graceful degradation under sensor failure, power loss, or GPS disruption). The craft uses a single servo-driven sail for both propulsion and steering against a static rudder, eliminating a conventional windvane in favour of a novel luff-detection approach described below.

Sensor Suite & Fusion Strategy

Sensor	Part / Module	Bus	Primary Outputs
Microcontroller	Seeed XIAO ESP32-S3	—	Main autonomy processor; I2C master, UART host, PWM servo output
9-Axis IMU	ICM-20948 (SparkFun breakout)	I2C 0x68	Gyro yaw rate, accelerometer (3-axis), magnetometer heading
Rotary Encoder	AS5600 + diametric magnet	I2C 0x36	Sail angle (12-bit, 0–360 deg), luff vibration jitter
GNSS / Compass	Seeed XIAO GNSS — Quectel L76K	UART	Lat/lon, COG, speed over ground; GPS + GLONASS + BeiDou + Galileo

Sensor Fusion Layers

Derived Quantity	Sensors Fused	Update Rate
Fast stable heading	ICM-20948 gyro + magnetometer (complementary filter)	50 Hz
Tilt-compensated heading	ICM-20948 magnetometer + accelerometer	50 Hz
Apparent wind direction	AS5600 sail angle + fused heading	On sweep
Leeway / spoofing flag	GPS COG vs compass heading	1 Hz
Dead-reckoning position	Gyro + accelerometer integrated	GPS fallback
Luff point / wind angle	AS5600 jitter variance during sweep	~every 10 min

Complementary filter formula (runs every IMU sample):

$$\text{heading} = 0.98 \times (\text{heading} + \text{gyro_yaw_rate} \times \text{dt}) + 0.02 \times \text{magnetometer_heading}$$

This blends the gyroscope's high-speed response with the magnetometer's long-term stability, delivering a heading estimate that is both fast enough to arrest a spin and drift-free over the course of a multi-day crossing.

Autopilot Control Loop

The main autopilot loop runs on the Seeed XIAO ESP32-S3, cycling approximately **every 200–500 milliseconds** (120–300 cycles per minute) — well matched to the slow dynamics of a sailing hull. Each cycle executes the following six steps:

#	Step	Detail
1	Read Sensors	Fused heading from ICM-20948 complementary filter; GPS position and COG from Quectel L76K via UART; AS5600 sail angle.
2	Compute Desired Heading	Great-circle bearing computed from current GPS lat/lon to the active waypoint coordinates.
3	Heading Error	$\text{error} = \text{desired_bearing} - \text{current_heading}$, normalised to ± 180 deg. Positive error means boat is left of course; negative means right.

4	Sail / Rudder Control	PID controller on heading error drives the MG996R sail servo. Gains tuned empirically during tethered trials. Sail angle limited to physical stop angles. ICM-20948 gyro yaw-rate inner loop at 50 Hz arrests spin onset before the outer navigation loop sees it.
5	Sail Trim Mode	CRUISE: small positive angle of attack for propulsion. FEATHER: sail at 0 deg angle of attack, fully depowered — used in high wind, low battery, or SAFE state. Mode selected from wind angle estimate and battery SoC reading.
6	Output + Repeat	PWM command sent to MG996R servo. Cross-track error adds a small bearing bias proportional to lateral offset from the direct track line. Loop repeats on the next 200–500 ms cycle.

Luff-Sweep Wind Angle Detection (No Windvane Required)

Conventional autonomous sailboats rely on a dedicated windvane to measure apparent wind angle. Olive Turtle eliminates this component by exploiting the **luffing phenomenon** — the characteristic flutter of a sail's leading edge as it crosses through the wind direction — detected via the AS5600 rotary encoder already fitted to the sail shaft.

What is Luffing?

As the sail rotates toward the wind direction its load transitions from a steady fill (smooth torque) to an unstable flutter (oscillating torque). This vibration travels back to the sail shaft, where the AS5600 reads it as high-frequency jitter in the position signal. Because luffing is symmetrical — occurring at equal angles either side of true wind — the midpoint between the two luff events precisely identifies apparent wind direction relative to the boat.

The Sweep Algorithm

Step	Action	Sensing
1	Slowly sweep sail toward wind at ~5–10 deg/sec	—
2	Compute rolling variance of AS5600 position over 200 ms window	Continuous
3	Record Angle A when variance exceeds luff threshold	Luff onset detected
4	Reverse sweep — back through filled sail, continue to opposite side	—
5	Record Angle B when variance exceeds threshold again	Second luff detected
6	$\text{wind_angle} = \text{midpoint}(\text{Angle A}, \text{Angle B})$	Wind angle solved
7	Return sail to optimal trim; store wind_angle for control loop	Resume navigation

- **Sweep cadence:** approximately every 10 minutes; more frequently after a tack or heading change greater than 30 deg.
- **First fix on boot:** full wide-arc sweep in ACQUIRE state to establish initial wind angle before SAIL-NAV is entered.
- **Light-wind adaptation:** variance threshold scaled down dynamically; sweep speed reduced to give the sail time to respond.
- **No-go zone logic:** if wind_angle places destination within 45 deg of upwind, autopilot initiates a tack sequence rather than pointing into the unworkable zone.

Waypoint Sequencing

Before deployment the operator loads a sequence of GPS waypoints (lat/lon). The sequencer starts at waypoint 1 and advances automatically once Olive Turtle enters the **arrival radius** (default 300 m). On reaching the final waypoint the system transitions to ARRIVAL. The sequencer always advances forward — it never doubles back. Cross-track bias in the control loop prevents the craft drifting far from the planned track line between waypoints.

Autonomous State Machine

What is a state machine? A state machine is a software design pattern that defines a fixed set of operating modes ('states') and the explicit rules for moving between them — ensuring the system always behaves predictably and safely, no matter what unexpected conditions arise.

Olive Turtle's five-state machine governs all high-level behaviour. On any reboot the system re-enters BOOT, ensuring the sail remains feathered until sensors are verified.

BOOT	Power-up initialisation. Hardware checked; awaiting valid GPS fix from the Quectel L76K and stable compass from the ICM-20948. Sail held feathered, rudder centred. No motion until all sensors verified.
ACQUIRE	Sensors ready; craft in standby near launch point. Sail feathered, boat drifts. Awaits operator start command. Full luff-sweep executed here to establish initial wind angle before SAIL-NAV is entered.
SAIL-NAV	Primary mission state. Autopilot loop active on XIAO ESP32-S3: PID heading control cycling every 200–500 ms, luff-sweep wind calibration every ~10 minutes, waypoint sequencing, and cross-track correction. Checks for arrival, faults, or geofence breaches each cycle.
ARRIVAL	Final waypoint reached. Sail feathered; rudder centred. Craft either drifts to beach naturally or enters a slow loiter within a defined geofence. Telemetry transmits a mission-complete notification if comms are available.
SAFE (Failsafe)	Catch-all safety state triggered by: extended GPS loss, critically low battery, watchdog reset, or geofence breach. Sail feathered to zero thrust; slow loiter or drift. Re-attempts GPS acquisition periodically. Requires manual reset to resume.
Transition flow: BOOT → ACQUIRE → SAIL-NAV → ARRIVAL (normal mission). Any state → SAFE (fault detected). Any reboot → BOOT (deterministic recovery).	

Reliability Features

Battery Failsafe

Voltage and SoC monitored continuously by the XIAO ESP32-S3. Below 20% capacity the autopilot feathers the sail automatically, conserving power for sensors and telemetry. Normal navigation resumes once solar recharge brings the battery above the safe threshold.

GPS Loss & Dead Reckoning

If no valid fix is received from the Quectel L76K for 60 seconds, Olive Turtle switches to dead-reckoning using ICM-20948 gyro and accelerometer integration, providing a short-duration position estimate while awaiting recovery. If GPS remains absent beyond a second threshold the craft enters SAFE mode and heaves-to.

GPS Spoofing Detection

GPS COG from the L76K is continuously compared with compass heading from the ICM-20948. Significant sustained disagreement beyond expected leeway flags a possible spoofing event and triggers SAFE. The L76K's multi-constellation support (GPS + GLONASS + BeiDou + Galileo) materially raises the spoofing difficulty threshold.

Software Watchdog

A background watchdog task on the ESP32-S3 monitors the main loop heartbeat. If the loop stalls across multiple cycles the MCU is reset. On reboot the system reads the last persisted waypoint index from non-volatile flash and resumes the mission via BOOT → ACQUIRE → SAIL-NAV without operator intervention.

Hardware Watchdog

The XIAO ESP32-S3 hardware watchdog timer is enabled independently of software. A dedicated pin must be toggled by the main loop each cycle; failure causes a hard reset. Combined with the state machine's deterministic boot sequence this ensures Olive Turtle cannot be left in a powered-but-unresponsive state.

Thermal & Environmental

If an internal temperature sensor detects overheating, or a moisture sensor detects water ingress, the system enters SAFE mode and broadcasts a distress telemetry packet if comms are available. The MG996R servo-driven sail will weathercock passively if power is lost, inherently depowering the rig.

Next Steps — Deploying the State Machine into the Olive Turtle Codebase

The hardware platform is confirmed operational: the Seeed XIAO ESP32-S3 is running, all sensors (ICM-20948, AS5600, Quectel L76K GNSS) are wired to the I2C and UART buses and returning data. The immediate priority is to layer autonomy logic on top of this verified sensor stack in six phases:

Phase 1 — Sensor Fusion

Implement the complementary filter for heading. Validate tilt compensation. Log fused heading vs raw magnetometer over a 10-minute bench test. Target: heading drift under 2 deg/min with no gimbal-lock artefacts on 30 deg roll.

Phase 2 — AS5600 Baseline

Confirm full 360 deg sail rotation maps cleanly to AS5600 counts. Write the rolling-variance function. Manually excite the sail to luff and verify the jitter spike exceeds 5x the calm baseline.

Phase 3 — BOOT / ACQUIRE States

Implement the state enum and transition logic. BOOT holds until the L76K provides a valid GPS fix and the ICM-20948 heading is stable. ACQUIRE then executes the first luff sweep and logs the initial wind_angle.

Phase 4 — SAIL-NAV + PID

Implement the PID heading controller driving the MG996R servo. Load a simple two-waypoint test mission and conduct a tethered water trial. Target: heading hold within +/-10 deg in calm water, advancing to WP2 on entering the arrival radius of WP1.

Phase 5 — ARRIVAL + SAFE States

Implement ARRIVAL (feather and loiter) and SAFE (feather, drift, periodic GPS retry). Simulate GPS dropout and low battery in a bench test. Target: SAFE entered within 60 s of GPS loss; sail feathered within 1 s; mission resumes after GPS mock-restore.

Phase 6 — Watchdog + Persistence

Enable the ESP32-S3 hardware watchdog. Persist the active waypoint index to flash. Force a reboot mid-mission in a bench test. Target: mission resumes from the correct waypoint; watchdog triggers a reset if the main loop is artificially stalled.